

Figure 3: Booting the installed operating system

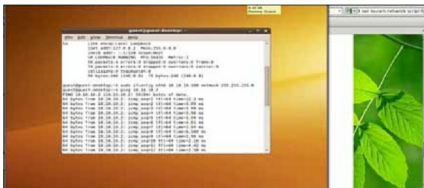


Figure 4: QEMU networking using kernel virtual device

device/interface `tap0`. In the next few steps, a bridge is created with `eth1` and `tap0` as parts of this bridge. Finally, we assign an IP address to this bridge:

```
#qemu-ifdown
ifconfig eth1 down
ifconfig br0 down
brctl delbr br0
ifconfig eth1 -promisc
ifup eth1
openvpn --rmtun --dev tap0
```

This script will be used to shut down QEMU networking; it is almost self-explanatory, shutting down both the interfaces, deleting the bridge, and the `tap0` device.

Copy these two files to your `/etc` directory, and test them:

```
# /etc/qemu-ifup
Wed Apr 6 15:53:50 2011 TUN/TAP device
tap0 opened
Wed Apr 6 15:53:50 2011 Persist state set
to: ON
# ifconfig br0
br0      Link encap:Ethernet Hwaddr
```

```
TX packets:0 errors:0 dropped:0
overruns:0 carrier:0
collisions:0 txqueuelen:100
RX bytes:0 (0.0 b) TX bytes:0
(0.0 b)
```

The virtual device `tap0` and bridge `br0` are up, so our script is working fine. Now bring it down, using the `qemu-ifdown` script:

```
# /etc/qemu-ifdown
Error: Connection activation failed:
Device not managed by NetworkManager
Wed Apr 6 15:56:44 2011 TUN/TAP device
tap0 opened
Wed Apr 6 15:56:44 2011 Persist state set
to: OFF
```

Every thing is set up correctly; now it's time to boot the Ubuntu VM with full networking support. Start the networking (as `root`), and boot the VM (as an ordinary user):

```
# /etc/qemu-ifup
$ qemu -m 512 -hda ubuntu.img -net nic
-net tap,ifname=tap0,script=no
```

When the machine boots up, assign an IP address to the `eth0` interface inside the VM:

```
$ sudo ifconfig eth0 10.10.10.100 netmask
255.255.255.0
```

Now try to ping the bridge IP (results are shown in Figure 4):

```
$ ping 10.10.10.2
```

The network is working fine in the VM, now experiment with it. Try setting up a proxy server on the host OS, and connect to the Internet from the guest OS.

This was just an introduction to QEMU. You can look forward to a lot more in subsequent articles. **END** 

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